

Solving Two-Step Equations

Worksheet 7A.01.03

Name _____

Date _____

Period _____

1. Match each property of equality to the example that shows it. Write the letter of the example on the line.

Every example below starts from $5 = 3 + 2$.

Property of Equality	Letter
Symmetric Property of Equality	
Addition Property of Equality	
Subtraction Property of Equality	
Multiplication Property of Equality	
Division Property of Equality	

A.	$5 \times 4 = (3 + 2) \times 4$
B.	$3 + 2 = 5$
C.	$5 \div 4 = (3 + 2) \div 4$
D.	$5 + 4 = 3 + 2 + 4$
E.	$5 - 4 = 3 + 2 - 4$

Use the equation $2 + 3n = 11$ for questions 2 through 5.

2. Complete the sentence to describe the equation.

The variable n is first being _____ by 3, and then 2 is being
_____ to this product. The value of the expression $2 + 3n$ is
_____ to 11.

3. Describe the operation that isolates $3n$ on the left side.
4. Describe the operation that isolates n on the left side.
5. Now solve $2 + 3n = 11$. Then check your solution by substitution.

For questions 6 through 11, solve the equation and check your solution by substitution.

6. $2x - 5 = 11$

7. $\frac{n}{12} - 3 = 7$

8. $6y + 5 = 29$

9. $9a + 7 = 88$

10. $\frac{y}{4} - 8 = 4$

11. $-8 + \frac{b}{3} = 10$